

Rational logarithms and irrationality exponent two

A deduction from the separated-weight multiplicity estimate

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Abstract

We give a proof draft of the extension $\mu(\log a) = 2$ for every positive rational $a \neq 1$, using the weighted transverse multiplicity estimate of OpenAI family 017 [1, Lemma 2.2] as the specialized external input. The logarithmic interpolation argument extends to points with distinct nonzero multiplicative coordinates; we supply that extension in full. For $a = A_0/B_0$, the additional column denominator $B_0^{(K-1)h}$ fits inside the original weighted degree budget as soon as $w_0 \geq (K-1)\log B_0$. The analytic determinant argument uses a Cauchy radius $4K(1 + |\log a|)$. Both requirements are compatible with the many-variable parameter limit. All interpolation parameters and approximation denominators are fixed before the degree tends to infinity.

1 Statement, status, and dependence

All logarithms of positive real numbers in this note are natural. For an irrational real number x , write

$$\mu(x) = \sup \left\{ \nu > 0 : 0 < \left| x - \frac{p}{q} \right| < q^{-\nu} \text{ for infinitely many coprime } p \in \mathbb{Z}, q \geq 2 \right\}.$$

Theorem 1.1 (Deduction from the multiplicity estimate in family 017). *Use the weighted transverse multiplicity estimate stated in Section 2, with its uniformity in the component and the analytic frame, as an input. If $a \in \mathbb{Q}_{>0} \setminus \{1\}$, then, for each real $\nu > 2$, there exists $Q(a, \nu)$ such that*

$$\left| \log a - \frac{p}{q} \right| \geq q^{-\nu} \quad (p \in \mathbb{Z}, q \in \mathbb{Z}, q \geq Q(a, \nu)).$$

The fractions in this assertion need not be reduced. In particular, $\log a$ is irrational and $\mu(\log a) = 2$.

Proof status. This is a research draft relative to the stated input from [1]. The distinct-center interpolation theorem, the arithmetic denominator accounting, the analytic determinant estimates, and the order of parameter choices are proved below. The role of the source input is explicit: the conclusion has the same dependence on that multiplicity estimate. No claim about independent external verification of the source manuscript is part of this note.

What changes from the period construction. Fix $\omega = \log a$. At the exact points $z = j\omega$ one has $e^z = a^j$, so the appropriate nearby rational centers are

$$(a^j, jp_1/q_1, \dots, jp_m/q_m), \quad 0 \leq j < K.$$

The identity

$$e^{j\omega + \log(1+t)} = a^j(1+t)$$

preserves the entire functions that generate the determinant cancellation. Since the numbers a^j are pairwise distinct, a curve with constant multiplicative coordinate can meet at most one center. This removes the extra argument needed for several centers in one multiplicative fiber.

The conclusion is pointwise in the fixed rational number a . The threshold $Q(a, \nu)$ obtained here is not effective. The proof establishes the exponent statement for every $\nu > 2$; it makes no assertion of a uniform positive constant in a q^{-2} lower bound.

2 Interpolation on distinct logarithmic fibers

For a positive rational vector $W = (w_0, \dots, w_m)$, let $P_W(H)$ be the space of polynomials in $Y, X_1, \dots, X_m$ spanned by the monomials $Y^h X_1^{\alpha_1} \dots X_m^{\alpha_m}$ satisfying $w_0 h + \sum_i w_i \alpha_i \leq H$. For another positive rational vector $V = (v_0, \dots, v_m)$, put

$$J_V(H) = \mathbb{C}[[t, u_1, \dots, u_m]] / (t^s u^\beta : v_0 s + \sum_i v_i \beta_i \geq H).$$

Thus $J_V(H)$ retains exactly the coefficients of weight strictly less than H . Throughout this section, $\log(1+t)$ denotes its formal power series at zero.

Theorem 2.1 (Interpolation at distinct multiplicative coordinates). *Fix integers $m, K \geq 1$ and positive rational numbers w_0, v_0, θ such that*

$$0 < \theta < 1, \quad K \frac{w_0}{v_0} \theta^m < 1. \quad (2.1)$$

There are successive lower thresholds for positive rational numbers $w_1, \dots, w_m$, depending only on these fixed data and the previously chosen weights, with the following property. Set

$$W = (w_0, w_1, \dots, w_m), \quad V = (v_0, w_1/\theta, \dots, w_m/\theta).$$

For arbitrary pairwise distinct $y_0, \dots, y_{K-1} \in \mathbb{C}^$ and arbitrary $c_{ji} \in \mathbb{C}$, the map*

$$\begin{aligned} P_W(H) &\longrightarrow \bigoplus_{j=0}^{K-1} J_V(H), \\ P &\longmapsto \left(P(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m) \right)_{j=0}^{K-1} \end{aligned} \quad (2.2)$$

is surjective for all sufficiently large, sufficiently divisible positive integers H . The divisibility requirement depends only on the weights; the remaining lower threshold for H may depend on the centers.

The weights are chosen before the centers. In particular, the theorem applies to $y_j = a^j$ for every fixed positive real $a \neq 1$, without any distinctness requirement on the additive coordinates c_{ji} .

The only specialized input is the weighted transverse multiplicity lemma from the proof for π . We state its precise form to make the dependence explicit. A *normal basis* is a set of coordinate or vector-field directions whose images form a basis of the ambient tangent space modulo the tangent space of the subvariety, at a general smooth point.

Lemma 2.2 (Weighted transverse multiplicity, input). *Give the coordinates $x_1, \dots, x_d$ positive rational degree weights $\rho_1, \dots, \rho_d$. On an open subset of $\mathbb{C}^d$, let $D_1, \dots, D_d$ be commuting analytic vector fields forming a frame, with positive rational costs $\kappa_1, \dots, \kappa_d$. Let $N > 0$ and*

$\epsilon > 0$. Suppose that polynomials $f_1, \dots, f_r$ have weighted degree at most N , that Z is an irreducible component of their common zero set of codimension $k \geq 1$, and that

$$D^\gamma f_\ell|_Z = 0 \quad \text{whenever} \quad \sum_b \kappa_b \gamma_b < \epsilon N.$$

For every coordinate normal basis A and every D -field normal basis B , each of cardinality k ,

$$\prod_{a \in A} \rho_a \leq 2 k! \epsilon^{-k} \prod_{b \in B} \kappa_b. \quad (2.3)$$

The bound is uniform in the equations, the component, and the analytic frame, with no lower threshold for N .

This is [1, Lemma 2.2]. In particular, its statement imposes no restriction that the multiplicative coordinate of an interpolation center equal 1.

2.1 Choosing the weights independently of the centers

Choose a positive rational σ so small that

$$(1 + 3\sigma)^{m+1} K(w_0/v_0) \theta^m < 1, \quad (1 + \sigma) \theta < 1. \quad (2.4)$$

Put $L_0 = m + 2$ and

$$C_{\text{sep}} = \max_{1 \leq k \leq m} 2 k! (L_0/\sigma)^k.$$

Write $v_i = w_i/\theta$ for $i > 0$. Choose the positive weights successively so that, for all subsets $A, B \subseteq \{0, \dots, m\}$ of equal cardinality between 1 and m ,

$$\prod_{a \in A} w_a > C_{\text{sep}} \prod_{b \in B} v_b \quad (2.5)$$

whenever the largest positive index in $A \triangle B$ belongs to A . No condition is imposed if there is no positive differing index.

These are successive lower-threshold conditions. If that largest index is $i \in A$, the ratio of the two products in (2.5) is

$$w_i \frac{w_0^{\mathbf{1}_{0 \in A}}}{v_0^{\mathbf{1}_{0 \in B}}} \theta^{|B \cap \{1, \dots, m\}|} \prod_{k < i} w_k^{\mathbf{1}_{k \in A} - \mathbf{1}_{k \in B}}.$$

It is w_i times a positive expression in earlier weights and the fixed data. All larger positive weights have canceled. There are finitely many conditions at each step, and subsequent choices do not change the ones already imposed. This construction uses no center coordinates.

2.2 The curve inequality

For an integral affine algebraic curve C , use its smooth projective model to define

$$\deg_W C = \sum_P \max \left\{ 0, -\frac{\text{ord}_P Y}{w_0}, -\frac{\text{ord}_P X_1}{w_1}, \dots, -\frac{\text{ord}_P X_m}{w_m} \right\}. \quad (2.6)$$

An identically zero coordinate contributes no pole. Set $a_j = (y_j, c_{j1}, \dots, c_{jm})$. At a branch P above a_j , put

$$h_P = \min \left\{ \frac{\text{ord}_P(Y/y_j - 1)}{v_0}, \left(\frac{\text{ord}_P(X_i - c_{ji} - \log(Y/y_j))}{v_i} \right)_{i=1}^m \right\}. \quad (2.7)$$

Here the logarithm is its formal expansion at $Y/y_j = 1$. These local coordinates have positive order, possibly infinite, and their minimum is finite on a nonconstant curve.

Proposition 2.3. *For the chosen weights, every such curve satisfies*

$$\deg_W C \geq (1 + \sigma) \sum_{P \rightarrow a_j} h_P, \quad (2.8)$$

where all branches through all prescribed centers are included.

Proof. There is nothing to prove if C misses the centers. Suppose that (2.8) fails. For every sufficiently large auxiliary integer N , linear algebra supplies a nonzero polynomial F_N of W -degree at most N whose local expansions at all centers have V -order at least $(1 + 3\sigma)N$. Indeed, the respective leading terms for its coefficient count and the imposed vanishing conditions are

$$\frac{N^{m+1}}{(m+1)! \prod_{a=0}^m w_a}, \quad \frac{K(1+3\sigma)^{m+1} N^{m+1}}{(m+1)! \prod_{a=0}^m v_a}.$$

Their comparison is the first inequality in (2.4). No independence among the vanishing conditions is needed.

On $U = \{Y \neq 0\}$, consider the commuting frame

$$D_0 = Y \partial_Y + \sum_{i=1}^m \partial_{X_i}, \quad D_i = \partial_{X_i} \quad (1 \leq i \leq m). \quad (2.9)$$

These vector fields preserve polynomial W -degree at most N . In the formal coordinates

$$t = Y/y_j - 1, \quad u_i = X_i - c_{ji} - \log(1+t)$$

at any center, they are $D_0 = (1+t)\partial_t$ and $D_i = \partial_{u_i}$. Differentiation by D_a lowers weighted order by at most v_a . Thus every $D^\gamma F_N$ with $\sum_a v_a \gamma_a \leq \sigma N$ has local order at least $(1 + 2\sigma)N$ at every center. On a branch P , its zero order is at least $(1 + 2\sigma)N h_P$.

Its restriction to the complete model of C has total pole degree at most $N \deg_W C$. The assumed failure of (2.8) makes the indicated total zero order larger than that pole bound. Hence

$$D^\gamma F_N|_C = 0 \quad \left(\sum_a v_a \gamma_a \leq \sigma N \right). \quad (2.10)$$

The order assertion uses only formal power series: every monomial has order at least its V -weight times h_P , and cancellation cannot lower the order.

Set $\delta = \sigma N / L_0$. For $0 \leq r \leq L_0$, let $\mathcal{V}_r \subseteq U$ be the common zero set of derivatives of cost at most $r\delta$. Starting from a component of $\mathcal{V}_{L_0}$ containing $C \cap U$, extend backwards to nested irreducible components of the earlier $\mathcal{V}_r$. Their dimensions lie between 1 and m : they contain a curve and lie in the proper zero set of F_N . Since there are $L_0 + 1 = m + 3$ levels, two adjacent components coincide. Denote their common value by Z , of codimension k .

Use the derivatives defining the earlier level as equations for Z . Every further derivative of cost less than δ vanishes on Z . Lemma 2.2, with $\epsilon = \sigma / L_0$, gives

$$\prod_{a \in A} w_a \leq C_{\text{sep}} \prod_{b \in B} v_b \quad (2.11)$$

for every coordinate normal basis A and frame normal basis B to Z . All normal-basis conditions may be checked on one common smooth open subset. The uniformity of the input permits the equations and Z to depend on N .

We claim that Y is constant on Z . If a positive coordinate direction is nontangent, choose the largest such index i . Every direction of larger positive index is tangent and therefore occurs

in neither kind of normal basis. Some coordinate normal basis A contains i . If a frame normal basis B omitted it, the largest positive differing index would belong to A , contradicting (2.5) and (2.11). Consequently every frame normal basis contains D_i .

The other frame vectors span the ambient hyperplane $\ker(dX_i - dY/Y)$. Their images fail to span the normal quotient, which is equivalent to the tangent space of Z being contained in that hyperplane. Thus

$$dX_i = dY/Y \quad \text{on } Z. \quad (2.12)$$

If Y were nonconstant, intersecting with general algebraic hyperplanes through a smooth point where $dY \neq 0$ would give an algebraic curve in Z on which Y is nonconstant. On its smooth complete model, dY/Y has a nonzero residue at a zero or pole of Y , whereas the differential of the rational function X_i has residue zero everywhere. This contradicts (2.12).

If all positive coordinate directions are tangent instead, these m independent directions exhaust the tangent space of the proper subvariety Z . Again $dY = 0$, so Y is constant in characteristic zero. The claim follows in either case.

It follows that C is contained in a fiber $Y = y_j$. Since the y_j are pairwise distinct, C meets at most one prescribed center. Choose X_l nonconstant on C . In this fiber $\log(Y/y_j) = 0$, and (2.7) implies $\text{ord}_P(X_l - c_{jl}) \geq v_l h_P$ at every branch through the center. The total pole degree of this nonzero rational function is at most $w_l \deg_W C$. Equality of its total zero and pole degrees gives

$$\deg_W C \geq \frac{v_l}{w_l} \sum_{P \rightarrow a_j} h_P = \theta^{-1} \sum_{P \rightarrow a_j} h_P > (1 + \sigma) \sum_{P \rightarrow a_j} h_P.$$

This contradicts the assumed failure and proves the proposition. $\square$

2.3 Realizing the coefficient packets

Proof of Theorem 2.1. Choose a sufficiently large positive integer R_0 such that R_0/w_a and R_0/v_a are positive integers for every a . Embed affine space by all monomials of W -degree at most R_0 , including the constant monomial and each coordinate. Let X be the projective closure and $L = \mathcal{O}_X(1)$. The chart of the constant monomial is exactly the original affine space; X is integral and projective. For a curve meeting this chart,

$$\deg(L|_C) = R_0 \deg_W C. \quad (2.13)$$

Indeed, on its smooth complete model, each embedding monomial has pole order at most R_0 times the maximum in (2.6). A pure coordinate power attains the maximum when it is positive, while the constant monomial attains it when it is zero. Clearing those maximal poles gives sections without a common zero and proves (2.13).

To make the local conditions algebraic, choose a Taylor polynomial $G_i(t)$ of $\log(1 + t)$ whose omitted monomials have t -weight strictly greater than v_i . At a_j use the regular local coordinates

$$t_j = Y/y_j - 1, \quad u'_{ji} = X_i - c_{ji} - G_i(t_j).$$

This replacement preserves h_P . If t_j is not identically zero on the branch, every omitted term has order divided by v_i strictly greater than $\text{ord}_P(t_j)/v_0$; if it is identically zero, the tail vanishes. The formal change from u_i to u'_i fixes t and adds only terms of weight greater than v_i to u_i . It and its inverse preserve the weighted filtration, so they induce inverse triangular maps on every desired coefficient packet.

At each center form the finite-colength ideal

$$I_j = (t_j^{R_0/v_0}, (u'_{j1})^{R_0/v_1}, \dots, (u'_{jm})^{R_0/v_m}). \quad (2.14)$$

These local ideals, with the unit ideal off the finite set of centers, define a coherent nonzero ideal sheaf I on X . For example, take the kernel of the map to the finite sum of their local Artinian quotients. At every branch through a center,

$$\mathrm{ord}_P I = R_0 h_P. \quad (2.15)$$

Let $p : X' \rightarrow X$ be the ordinary blowup of I , with $I\mathcal{O}_{X'} = \mathcal{O}_{X'}(-E)$. Then E is an effective Cartier divisor and $-E$ is relatively ample. Write $A = p^*L$ in additive notation. We claim that

$$A - (1 + \sigma)E \quad \text{is nef.} \quad (2.16)$$

For a noncontracted curve Γ upstairs whose image meets the affine chart, the map to its image is birational, since the blowup is an isomorphism away from finitely many centers. On complete normalizations, (2.13) and (2.15) give

$$A \cdot \Gamma = R_0 \deg_W p(\Gamma), \quad E \cdot \Gamma = R_0 \sum_{P \mapsto a_j} h_P.$$

Proposition 2.3 gives the required nonnegative degree. A curve whose image misses the affine chart avoids E . A contracted curve has A -degree zero and positive $-E$ -degree by relative ampleness. These cases account for all curves and prove (2.16).

For a sufficiently large integer $b > 1$, the divisor $bA - E$ is ample: a relatively ample line bundle becomes ample after tensoring with a sufficiently large pullback power of L . Now

$$A - E = \frac{b-1}{b(1+\sigma)-1}(A - (1+\sigma)E) + \frac{\sigma}{b(1+\sigma)-1}(bA - E).$$

It is a positive combination of a nef and an ample class, hence ample. These statements apply to projective schemes; no smoothness or normality of the blowup is required.

For all sufficiently large n , the eventual graded-piece theorem for the ordinary Rees algebra and relative Serre vanishing give

$$p_*\mathcal{O}_{X'}(-nE) = I^n, \quad R^q p_*\mathcal{O}_{X'}(-nE) = 0 \quad (q > 0). \quad (2.17)$$

On an affine open in the Noetherian base, its Rees algebra is generated in degree one, its sufficiently high graded pieces recover the sections on relative Proj, and the higher cohomology of those twists vanishes. A finite affine cover gives a common threshold in (2.17). These are ordinary powers I^n , with no assertion for small n and no replacement by integral closures.

Projection formula, Leray, and Serre vanishing for the ample line bundle $A - E$ therefore yield

$$H^1(X, I^n \otimes L^n) = H^1(X', \mathcal{O}_{X'}(n(A - E))) = 0$$

for sufficiently large n . The exact sequence for I^n shows that $H^0(X, L^n)$ surjects onto all local data modulo I^n , using the constant embedding coordinate to trivialize L on the affine chart. Finite-colength quotients are unchanged by completion.

Every displayed generator of I_j has V -weight R_0 . Consequently

$$I_j^n \subseteq \{\text{formal series of } V\text{-weight at least } nR_0\}. \quad (2.18)$$

Surjectivity modulo I_j^n thus implies surjectivity onto the smaller quotient $J_V(nR_0)$. Equality in (2.18) is not needed.

Finally, for sufficiently large n , each section of L^n is the restriction of a degree- n homogeneous polynomial in the embedding coordinates. This follows from Serre vanishing for the

homogeneous ideal sheaf of X in its ambient projective space. On the affine chart such a polynomial has W -degree at most nR_0 . Taking $H = nR_0$ and undoing the filtration-preserving Taylor-polynomial change proves (2.2).

The integer R_0 depends only on the weights. The subsequent lower threshold for n may depend on the centers through I , as allowed in the theorem. Together with the preceding center-independent weight choice, this proves the stated order of quantifiers. $\square$

Remark 2.4. The source theorem uses the additional condition $K\theta^m < 1$ to handle multiple centers in the same fiber $Y = 1$. Distinct y_j let the curve argument stop as soon as Y is constant, so that condition is not used here. The parameters in the determinant application may nevertheless be chosen to satisfy it as well. The extension above is proved from the multiplicity input and the curve-to-jet argument; it is not obtained by citing an unstated arbitrary- Y version of the source interpolation theorem.

3 Arithmetic and analytic determinant estimates

Fix coprime positive integers A_0, B_0 with $A_0 \neq B_0$, and put

$$a = \frac{A_0}{B_0}, \quad \omega = \log a \neq 0.$$

All logarithms are natural. In this section all parameters except the polynomial degree H are fixed. For a multi-index α write

$$w\alpha = \sum_{i=1}^m w_i \alpha_i, \quad |\alpha| = \sum_{i=1}^m \alpha_i, \quad w_* = \min_{1 \leq i \leq m} w_i.$$

Multi-indices have nonnegative integer entries, and inequalities between them are coordinate-wise. Suppose that $\nu > 2$ and that integers p_i and $q_i \geq 2$ satisfy

$$\left| \omega - \frac{p_i}{q_i} \right| \leq q_i^{-\nu}, \quad r_i = \frac{p_i}{q_i}, \quad w_i = \lceil \log q_i \rceil \quad (1 \leq i \leq m). \quad (3.1)$$

We assume that $m, K, w_0, v_0, \theta, w_1, \dots, w_m$ satisfy the hypotheses of Theorem 2.1, for the centers $(a^j, jr_1, \dots, jr_m)$, $0 \leq j < K$. In particular, w_0, v_0, θ are positive rational numbers and $0 < \theta < 1$. Besides the volume inequality required by that theorem, we additionally impose $K\theta^m < 1$. Thus our numerical assumptions are

$$K\theta^m < 1, \quad K \frac{w_0}{v_0} \theta^m < 1. \quad (3.2)$$

The additional restriction is harmless: all final choices satisfy both inequalities. The local parameter along the logarithmic curve is chosen so that its first coordinate is $a^j(1+t)$.

Choose a rational number $F_0 > 2/\theta$, and define

$$T_i = \left\lceil \frac{F_0 w_i}{v_0} \right\rceil, \quad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1} t^k}{k}. \quad (3.3)$$

An empty sum is zero. The first term omitted from G_i has weight $v_0 T_i \geq F_0 w_i > w_i/\theta$. Thus the substitution $u_i \mapsto u_i + G_i(t) - \log(1+t)$ is invertible and preserves the weighted filtration on the jet space. Replacing each logarithm in Theorem 2.1 by G_i therefore preserves surjectivity.

3.1 The matrix and its row count

Use as columns the monomials

$$P(Y, X) = Y^h X^\alpha, \quad h \in \mathbb{Z}_{\geq 0}, \quad \alpha \in \mathbb{Z}_{\geq 0}^m, \quad w_0 h + w\alpha \leq H,$$

and as rows the triples

$$\rho = (j, s, \beta), \quad 0 \leq j < K, \quad s \in \mathbb{Z}_{\geq 0}, \quad \beta \in \mathbb{Z}_{\geq 0}^m, \quad v_0 s + \frac{w\beta}{\theta} < H.$$

The entry in this row and column is

$$[t^s u^\beta] P(a^j(1+t), jr_1 + G_1(t) + u_1, \dots, jr_m + G_m(t) + u_m). \quad (3.4)$$

For all sufficiently large, sufficiently divisible integers H , Theorem 2.1 supplies a nonzero square minor Δ_H using every row. The matrix has rational entries, so the minor is a nonzero rational number. Its selected columns may depend on H . Let $M = M_H$ be its size, and set

$$\bar{b} = \bar{b}_H = \frac{1}{MH} \sum_{\rho=(j,s,\beta)} w\beta. \quad (3.5)$$

The sums over rows below include all their multiplicities.

The elementary lattice count for a simplex gives

$$M \sim \frac{K\theta^m H^{m+1}}{(m+1)!v_0 \prod_{i=1}^m w_i}, \quad 0 \leq \bar{b} \leq \theta. \quad (3.6)$$

Indeed, after division of the coordinates by H , the relevant region is the simplex $v_0 s + \sum_i w_i \beta_i / \theta < 1$. Its volume is $\theta^m / ((m+1)!v_0 \prod_i w_i)$, and its boundary has measure zero. Riemann sums therefore give the first assertion, including the strict inequality defining the rows. The second follows row by row from $w\beta < \theta H$. All weights are fixed before this limit is taken.

3.2 The arithmetic lower bound

The rational first coordinates introduce a column denominator. Their cost is paid by the same weighted degree budget as the X -coordinates; no average-degree assertion about the selected columns is required.

Lemma 3.1 (Arithmetic lower bound). *Suppose in addition that*

$$w_0 \geq (K-1) \log B_0. \quad (3.7)$$

Put $\Lambda = 4 \log 2$ and

$$E_{\text{ar}} = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_{i=1}^m \frac{1}{w_i} + \frac{\theta}{w_*}. \quad (3.8)$$

Then every nonzero minor selected above satisfies

$$\frac{\log |\Delta_H|}{MH} \geq -(1 - \bar{b}) - E_{\text{ar}}. \quad (3.9)$$

Proof. Let

$$L_i = \text{lcm}(1, \dots, T_i - 1), \quad D_H = \prod_{i=1}^m L_i^{\lfloor H/w_i \rfloor},$$

where the least common multiple of an empty list is 1. Multiply the column $Y^h X^\alpha$ by $B_0^{(K-1)h} \prod_i q_i^{\alpha_i}$ and the row (j, s, β) by $\prod_i q_i^{-\beta_i}$. The scaled entry is zero unless $\alpha \geq \beta$. Otherwise it equals

$$A_0^{jh} B_0^{(K-1-j)h} \left(\prod_i \binom{\alpha_i}{\beta_i} \right) [t^s] (1+t)^h \prod_i (jp_i + q_i G_i(t))^{\alpha_i - \beta_i}. \quad (3.10)$$

Both displayed powers of A_0, B_0 are nonnegative integer powers. Since $G_i \in L_i^{-1} \mathbb{Z}[t]$, the remaining denominator divides $\prod_i L_i^{\alpha_i - \beta_i}$, which divides D_H . Multiplying each entry of the scaled matrix by D_H consequently gives an integer matrix with nonzero determinant. Taking absolute values of its determinant yields

$$\begin{aligned} \log |\Delta_H| \geq & -M \log D_H - \sum_{\text{selected } (h, \alpha)} \left((K-1)h \log B_0 + \sum_i \alpha_i \log q_i \right) \\ & + \sum_{\text{rows } \beta} \sum_i \beta_i \log q_i. \end{aligned} \quad (3.11)$$

For every selected column, separately, its cost is at most

$$(K-1)h \log B_0 + w\alpha \leq w_0 h + w\alpha \leq H$$

by (3.7). Also $0 \leq w_i - \log q_i < 1$, so

$$\sum_{\text{rows } \beta} \sum_i \beta_i \log q_i \geq MH\bar{b} - \sum_{\text{rows } \beta} |\beta| \geq MH\bar{b} - \frac{MH\theta}{w_*}.$$

We use the elementary bound $\log \text{lcm}(1, \dots, n) \leq \Lambda n$. To verify it, the prime powers in $(\ell, 2\ell]$ contribute at most $\log \binom{2\ell}{\ell} \leq 2\ell \log 2$ to the increase of the logarithm of the least common multiple. Summing this inequality over dyadic intervals up to the smallest power of two at least n gives the stated bound with $\Lambda = 4 \log 2$. Thus

$$\frac{\log D_H}{H} \leq \Lambda \sum_i \frac{T_i}{w_i} \leq \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

Substituting these estimates in (3.11) proves the lemma. $\square$

3.3 Exact translation to entire functions

For every multi-index γ and selected column P , define

$$f_{\gamma, P}(z) = [u^\gamma] P(e^z, z + u_1, \dots, z + u_m). \quad (3.12)$$

For the column $P = Y^h X^\alpha$ this is the entire function

$$f_{\gamma, P}(z) = \begin{cases} \binom{\alpha}{\gamma} e^{hz} z^{|\alpha| - |\gamma|}, & \gamma \leq \alpha, \\ 0, & \text{otherwise.} \end{cases} \quad (3.13)$$

In particular only $w\gamma \leq H$ can contribute. For fixed γ these functions form a row vector indexed by the selected columns.

Lemma 3.2 (Exact row translation). *Every row $\rho = (j, s, \beta)$ of (3.4) is a sum of at most*

$$Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1) \quad (3.14)$$

scalar multiples of rows

$$([t^\ell] f_{\gamma, P}(j\omega + \log(1+t)))_P, \quad \gamma \geq \beta, \quad w\gamma \leq H, \quad 0 \leq \ell \leq s.$$

Each scalar ξ accompanying index γ satisfies

$$|\xi| \leq \exp\{-\nu w(\gamma - \beta) + HE_{\text{tr}}\}, \quad E_{\text{tr}} = \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log K + \nu}{w_*}. \quad (3.15)$$

Proof. Put

$$\epsilon_{ji} = j(r_i - \omega), \quad \tau_i(t) = G_i(t) - \log(1+t), \quad z_j(t) = j\omega + \log(1+t).$$

The equality $e^{z_j(t)} = a^j(1+t)$ identifies the first coordinate exactly. Expanding the other coordinates first in u , and then in the constants ϵ_{ji} and the tails τ_i , gives

$$\begin{aligned} \text{row}_{j,s,\beta} = & \sum_{\substack{\gamma \geq \beta \\ w\gamma \leq H}} \binom{\gamma}{\beta} \sum_{0 \leq d \leq \gamma - \beta} \binom{\gamma - \beta}{d} \epsilon_j^{\gamma - \beta - d} \sum_{k=0}^s \left([t^k] \prod_i \tau_i(t)^{d_i} \right) \\ & \cdot ([t^{s-k}] f_{\gamma, P}(z_j(t)))_P. \end{aligned} \quad (3.16)$$

Every scalar in this identity is independent of P , which is needed for the subsequent use of determinant multilinearity.

The product of tails vanishes to order at least $\sum_i d_i T_i$. Thus any nonzero coefficient in (3.16) satisfies

$$wd \leq \frac{v_0}{F_0} \sum_i d_i T_i \leq \frac{v_0 s}{F_0} < \frac{H}{F_0}. \quad (3.17)$$

On $|t| = 1/2$ one has

$$|\tau_i(t)| \leq \sum_{k \geq T_i} \frac{2^{-k}}{k} \leq 1.$$

Cauchy's coefficient estimate therefore bounds the coefficient of the tail product by $2^k \leq 2^s$. From (3.1),

$$|\epsilon_{ji}| \leq K q_i^{-\nu} \leq K e^\nu e^{-\nu w_i}.$$

Furthermore,

$$\binom{\gamma}{\beta} \binom{\gamma - \beta}{d} \leq 4^{|\gamma|}, \quad |\gamma| \leq H/w_*.$$

Combining these inequalities with (3.17), the logarithm of the modulus of any nonzero scalar is at most

$$-\nu w(\gamma - \beta) + \frac{\nu H}{F_0} + \frac{H \log 2}{v_0} + \frac{H(\log 4 + \log K + \nu)}{w_*}.$$

A zero scalar satisfies the same upper bound. Each coordinate of γ, d is at most $\lfloor H \rfloor$, since $w_i \geq 1$, and $0 \leq k \leq H/v_0$. This proves the term count and the lemma. $\square$

3.4 A collision bound for repeated transverse indices

Lemma 3.3 (Collision estimate). *Choose one translated test row from each row expansion in Lemma 3.2, omitting its scalar factor, and let T be the resulting square matrix. Let n_γ be the number of its rows with transverse index γ . Set*

$$R = 4K(1 + |\omega|), \quad \kappa = \frac{\log 2}{4}, \quad E_{\text{hol}} = \frac{R}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R)}{w_*}. \quad (3.18)$$

There is a quantity $\rho_H \rightarrow 0$, uniform over the choices of rows and selected columns, such that

$$|\det T| \leq \exp \left\{ -\kappa \sum_{\gamma} n_{\gamma}^2 + MH(E_{\text{hol}} + \rho_H) \right\}. \quad (3.19)$$

Proof. Equation (3.13) gives, on $|z| \leq R$,

$$|f_{\gamma,P}(z)| \leq \exp \left\{ H \left(\frac{R}{w_0} + \frac{\log(2R)}{w_*} \right) \right\} =: \mathcal{D}_H.$$

Indeed, $\binom{\alpha}{\gamma} \leq 2^{|\alpha|}$, $h \leq H/w_0$, $|\alpha| \leq H/w_*$, and $R \geq 1$. Write

$$f_{\gamma,P}(z) = \sum_{d \geq 0} c_{\gamma,d,P} (z/R)^d.$$

Cauchy's estimate on the circle of radius R yields $|c_{\gamma,d,P}| \leq \mathcal{D}_H$. On $|t| = 1/2$,

$$|j\omega + \log(1+t)| \leq (K-1)|\omega| + \log 2 < R/2.$$

The test of order ℓ at any of these centers consequently takes $(z/R)^d$ to a number of modulus at most $2^\ell 2^{-d}$.

Expand the determinant in these power-series rows. The expansion is absolutely convergent: each coefficient determinant is at most $M! \mathcal{D}_H^M$, and the independent geometric sums of 2^{-d} converge. If two rows have the same pair (γ, d) , their coefficient vectors $(c_{\gamma,d,P})_P$ coincide, and that determinant term vanishes. Thus every nonzero term has distinct degrees within each transverse-index group and satisfies

$$\sum_{\text{rows}} d \geq \sum_{\gamma} \binom{n_{\gamma}}{2}.$$

Use half of the geometric decay for this inequality, and sum the other half independently over all nonnegative degrees. Since $\sum_{\text{rows}} \ell \leq MH/v_0$ and $\sum_{\gamma} n_{\gamma} = M$, this gives

$$|\det T| \leq M! \mathcal{D}_H^M 2^{MH/v_0} 2^{-\frac{1}{4} \sum_{\gamma} n_{\gamma}^2 + M/4} (1 - 2^{-1/2})^{-M}. \quad (3.20)$$

Using $\log(M!) \leq M \log M$, one may take

$$\rho_H = \frac{\log M + \frac{1}{4} \log 2 - \log(1 - 2^{-1/2})}{H}.$$

Equation (3.6) implies $\rho_H \rightarrow 0$ with all parameters fixed. Its expression depends neither on the row choices nor on which full-row minor was selected. $\square$

3.5 The determinant comparison

Put $E_{\text{an}} = E_{\text{tr}} + E_{\text{hol}}$. A cutoff A_*H for the transverse weights leads to two alternatives. Many indices below the cutoff force repeated transverse indices and a collision saving; many indices above the cutoff instead force powers of the approximation errors.

Proposition 3.4 (Determinant comparison). *Let $0 < A_* < 1$ and $0 < \eta < 1$, and define*

$$g = \nu(A_*(1 - \eta) - \theta) - (1 - \theta), \quad \mathcal{L} = \frac{\eta^2 K \theta^m}{(m+1)v_0 A_*^m}. \quad (3.21)$$

The approximation and interpolation hypotheses, together with (3.7), cannot all hold if

$$g > 0, \quad E_{\text{ar}} + E_{\text{an}} < g, \quad \kappa \mathcal{L} > 1 + E_{\text{ar}} + E_{\text{an}}. \quad (3.22)$$

Proof. Let

$$D_{A_*}(H) = \#\{\gamma \in \mathbb{Z}_{\geq 0}^m : w\gamma \leq A_*H\}, \quad \mathcal{L}_H = \frac{\eta^2 M}{H D_{A_*}(H)}.$$

The simplex count in dimension m gives

$$D_{A_*}(H) \sim \frac{(A_*H)^m}{m! \prod_i w_i}, \quad \mathcal{L}_H \rightarrow \mathcal{L},$$

where the second assertion also uses (3.6).

Expand Δ_H by Lemma 3.2. For one choice of indices γ_ρ , its product of scalar factors has modulus at most

$$\exp \left\{ -\nu \sum_\rho w\gamma_\rho + \nu M H \bar{b} + M H E_{\text{tr}} \right\}. \quad (3.23)$$

There are at most Q_H^M choices. If at least ηM chosen indices satisfy $w\gamma_\rho \leq A_*H$, Cauchy–Schwarz gives

$$\sum_\gamma n_\gamma^2 \geq \frac{(\eta M)^2}{D_{A_*}(H)} = M H \mathcal{L}_H.$$

The main exponent $-\nu \sum_\rho w(\gamma_\rho - \beta_\rho)$ in (3.23) is nonpositive, since $\gamma_\rho \geq \beta_\rho$. Discarding it and applying Lemma 3.3 yields the normalized upper bound $-\kappa \mathcal{L}_H + E_{\text{an}} + \rho_H$ for that term.

Otherwise more than $(1 - \eta)M$ chosen indices have weight exceeding A_*H . Thus $\sum_\rho w\gamma_\rho \geq M H A_*(1 - \eta)$. Discard the nonpositive collision exponent and use (3.23); the normalized upper bound is $-\nu(A_*(1 - \eta) - \bar{b}) + E_{\text{an}} + \rho_H$. Taking absolute values before summing all choices therefore gives

$$\frac{\log |\Delta_H|}{M H} \leq E_{\text{an}} + \varepsilon_H + \max\{-\kappa \mathcal{L}_H, -\nu(A_*(1 - \eta) - \bar{b})\}, \quad (3.24)$$

where $\varepsilon_H = \rho_H + \log Q_H/H \rightarrow 0$.

For the second term in the maximum, its saving relative to $1 - \bar{b}$ is at least g , because $\bar{b} \leq \theta$ and

$$\nu(A_*(1 - \eta) - \bar{b}) - (1 - \bar{b}) = \nu A_*(1 - \eta) - 1 - (\nu - 1)\bar{b} \geq g.$$

For the first term use $1 - \bar{b} \leq 1$ and $\mathcal{L}_H \rightarrow \mathcal{L}$. Under (3.22), both alternatives in (3.24) are strictly smaller than $-(1 - \bar{b}) - E_{\text{ar}}$ for all sufficiently large permissible H . This contradicts Lemma 3.1. $\square$

4 Choice of parameters and the irrationality exponent

The dimension is chosen before the approximation denominators. The polynomial degree tends to infinity only after all these quantities and the interpolation centers have been fixed.

Lemma 4.1 (Choice of the dimension parameters). *For every real $\nu > 2$ there are positive rational numbers $\theta, A_*, B_*, C_*, \eta$ such that*

$$0 < \theta < A_* < B_* < 1, \quad C_* > 1, \quad 0 < \eta < 1, \quad (4.1)$$

$$C_* B_* < 1, \quad \frac{C_* \theta}{B_*} > 1, \quad C_* \theta < 1, \quad g = \nu(A_*(1 - \eta) - \theta) - (1 - \theta) > 0. \quad (4.2)$$

Proof. Choose a rational θ with $(\nu - 1)^{-2} < \theta < 1$. For $x = \sqrt{\theta}$,

$$(\nu - 1)x^2 - \nu x + 1 = (x - 1)((\nu - 1)x - 1) < 0.$$

Consequently

$$\theta + \frac{1 - \theta}{\nu} < \sqrt{\theta}.$$

Choose rational numbers satisfying

$$\theta + \frac{1 - \theta}{\nu} < A_* < B_* < \sqrt{\theta},$$

and then a rational C_* in the nonempty interval $(B_*/\theta, 1/B_*)$. Its nonemptiness follows from $B_*^2 < \theta$. Since $B_* > \theta$, these choices give $C_* > 1$ and $C_* \theta < \theta/B_* < 1$, as well as the other two required ratios. At $\eta = 0$ the expression defining g is positive. A sufficiently small positive rational η preserves this strict inequality. $\square$

4.1 Completion of the proof

Proof of Theorem 1.1. Write $a = A_0/B_0$ in lowest positive terms and fix $\nu > 2$. Suppose there are arbitrarily large positive integers q for which some $p \in \mathbb{Z}$ satisfies

$$\left| \omega - \frac{p}{q} \right| \leq q^{-\nu}, \quad \omega = \log a. \quad (4.3)$$

Choose $\theta, A_*, B_*, C_*, \eta, g$ as in Lemma 4.1, and put $\epsilon_0 = \min(g/2, 1/2)$. First choose a rational $F_0 > 2/\theta$ so large that $\nu/F_0 < \epsilon_0/3$.

For each positive integer m , set

$$K = \lfloor C_*^m \rfloor, \quad w_0 = B_*^{-m}, \quad v_0 = 2K\theta^m w_0. \quad (4.4)$$

These are permissible rational data for interpolation:

$$K\theta^m \leq (C_*\theta)^m < 1, \quad K \frac{w_0}{v_0} \theta^m = \frac{1}{2}.$$

As $m \rightarrow \infty$,

$$\frac{K}{w_0} \leq (C_* B_*)^m \rightarrow 0, \quad v_0 \sim 2 \left(\frac{C_* \theta}{B_*} \right)^m \rightarrow \infty, \quad \frac{m}{v_0} \rightarrow 0. \quad (4.5)$$

Moreover, the quantity in (3.21) satisfies

$$\mathcal{L} = \frac{\eta^2}{2(m+1)} \left(\frac{B_*}{A_*} \right)^m \rightarrow \infty. \quad (4.6)$$

Put $R = 4K(1 + |\omega|)$. Because a is fixed, (4.5) also gives

$$\frac{(K-1)\log B_0}{w_0} \longrightarrow 0, \quad \frac{R}{w_0} \longrightarrow 0.$$

We may therefore choose one finite m sufficiently large that

$$w_0 \geq (K-1)\log B_0, \quad \frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{R}{w_0} < \frac{\epsilon_0}{3}, \quad \kappa\mathcal{L} > 2. \quad (4.7)$$

Fix this m , and hence all parameters in (4.4) and the radius R .

Now choose approximations p_i/q_i satisfying (4.3), with $w_i = \lceil \log q_i \rceil$, successively large enough for Theorem 2.1 and also for

$$\Lambda \sum_{i=1}^m \frac{1}{w_i} + \frac{\theta + \log 4 + \log K + \nu + \log(2R)}{w_*} < \frac{\epsilon_0}{3}. \quad (4.8)$$

To do this, first impose a sufficiently large common lower bound on the w_i , using $\sum_i 1/w_i \leq m/w_*$. Then impose the successive interpolation thresholds. The denominators in (4.3) are unbounded, so all these finitely many requirements can be met. The thresholds do not depend on the as-yet unchosen rational centers. The first coordinates a^j are nonzero and pairwise distinct because $a > 0$ and $a \neq 1$. Theorem 2.1 therefore applies to the selected additive coordinates jr_i without any further restriction on them.

The sum of the three errors is exactly

$$\begin{aligned} E_{\text{ar}} + E_{\text{tr}} + E_{\text{hol}} &= \frac{\nu}{F_0} + \frac{\Lambda F_0 m + 2\log 2}{v_0} + \frac{R}{w_0} + \Lambda \sum_{i=1}^m \frac{1}{w_i} \\ &\quad + \frac{\theta + \log 4 + \log K + \nu + \log(2R)}{w_*}. \end{aligned} \quad (4.9)$$

Our choices make this less than ϵ_0 . It is therefore less than g , and $\kappa\mathcal{L} > 2 > 1 + E_{\text{ar}} + E_{\text{tr}} + E_{\text{hol}}$. All hypotheses of Proposition 3.4 hold, a contradiction. In that proposition the degree H tends to infinity only after the dimension, weights, approximations, and centers have been fixed.

This excludes (4.3) for unbounded q and in fact proves the strict eventual inequality

$$\left| \log a - \frac{p}{q} \right| > q^{-\nu} \quad (p \in \mathbb{Z}, q \in \mathbb{Z}, q \geq Q(a, \nu)). \quad (4.10)$$

The contradiction permitted unreduced fractions and zero approximation error. If ω were rational, multiplying its numerator and denominator by arbitrarily large integers would give zero-error instances of (4.3). Thus ω is irrational.

The upper bound for every $\nu > 2$ gives $\mu(\omega) \leq 2$. For the opposite inequality, partition $[0, 1)$ into N equal intervals and apply pigeonhole to the fractional parts of $0, \omega, \dots, N\omega$. There are integers p, q with $1 \leq q \leq N$ and

$$0 < \left| \omega - \frac{p}{q} \right| \leq \frac{1}{qN} \leq \frac{1}{q^2}.$$

Reducing the fraction preserves an upper bound by the reciprocal square of its new denominator. As $N \rightarrow \infty$, the reduced denominators cannot remain bounded, since ω is irrational and the approximation errors tend to zero. We obtain infinitely many reduced approximations of exponent at least 2, and therefore $\mu(\omega) \geq 2$. This proves the equality. $\square$

5 Where the value two enters

The compatible parameter region has a transparent description. Before allowing a small positive error fraction η , the two strict requirements on the cutoff A_* are

$$\nu(A_* - \theta) > 1 - \theta, \quad A_*^2 < \theta.$$

Thus it is enough to choose

$$\theta + \frac{1 - \theta}{\nu} < A_* < \sqrt{\theta}.$$

For $x = \sqrt{\theta} \in (0, 1)$, this interval is nonempty exactly when

$$(\nu - 1)x^2 - \nu x + 1 = (x - 1)((\nu - 1)x - 1) < 0,$$

or $1/(\nu - 1) < x < 1$. Such a choice exists for every $\nu > 2$. At $\nu = 2$, the arithmetic-geometric mean inequality gives $(1 + \theta)/2 \geq \sqrt{\theta}$, and this strict parameter interval is empty. The removal of the auxiliary same-fiber condition $K\theta^m < 1$ from the geometric theorem does not remove this balance between approximation gain and analytic growth.

Scope of the argument. Rationality of a is used precisely in clearing the determinant over $\mathbb{Q}$. For a general algebraic a , a norm would also involve the other embeddings, and the arithmetic comparison would require new estimates. That extension is not proved here.

References

- [1] OpenAI, *The irrationality exponent of π is 2*, 24 September 2026, family 017 of the `openai/math` repository. Specialized input: Lemma 2.2; source organization of the interpolation and determinant method: Sections 2–4. Public manuscript on GitHub.